

# Manual Winch

## Technical Datasheet

**Purpose of this datasheet**

Generic technical information for manual winches used in theatre, venue and production environments. It is intended to support the portal, inspection planning and site record keeping.

**Status**

Template document - verify against the actual manufacturer instructions, installed configuration and competent person requirements before use.

### 1. Equipment summary

A manual winch is a hand-operated lifting or pulling appliance used to raise, lower, tension or position a load using a rope, wire rope, belt or drum arrangement. In theatres and venues, manual winches may be used with curtains, scenery, bars, screens or other production equipment depending on the installed design.

The winch, handle, brake or locking device, drum, rope or wire rope, guides, pulleys, terminations, fixings and supporting structure should be considered as one complete load path.

**Important limitation**

This document is a generic technical datasheet. It does not replace the manufacturer manual, project design information, local risk assessment, written scheme of examination, or competent person advice.

## 2. Identification and site details

|                                                   |                                                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Asset / equipment name</b>                     | Manual Winch                                                                                      |
| <b>Venue / site</b>                               | To be completed                                                                                   |
| <b>Location / line number</b>                     | To be completed                                                                                   |
| <b>Manufacturer</b>                               | To be completed                                                                                   |
| <b>Model / type</b>                               | To be completed                                                                                   |
| <b>Serial / asset number</b>                      | To be completed                                                                                   |
| <b>SWL / WLL</b>                                  | To be confirmed from manufacturer / installation data                                             |
| <b>Rope / wire rope / belt type</b>               | To be completed                                                                                   |
| <b>Drum / winding arrangement</b>                 | To be completed                                                                                   |
| <b>Mounting / fixing method</b>                   | To be completed                                                                                   |
| <b>Operating method</b>                           | Manual handle / crank / geared operation - confirm exact arrangement                              |
| <b>Inspection / thorough examination interval</b> | To be confirmed by competent person or written scheme                                             |
| <b>Associated documents</b>                       | Manufacturer manual, installation drawings, certificates, inspection reports, maintenance records |

## 3. Intended use

- Designed for manual lifting, lowering, tensioning or positioning only where the installed design permits that use.
- Must only be used within the stated SWL / WLL, duty, travel and configuration limits.
- Must be operated by competent persons who understand the load, route of travel, stopping method and locking or braking arrangement.
- Must not be used for lifting people unless specifically designed, rated and assessed for that purpose.
- Must not be modified, drilled, welded, repaired or reconfigured without competent assessment and approval.

## 4. Key components and information to confirm

| Component / area               | Purpose                                               | Typical information to confirm                                                                |
|--------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Winch frame / body</b>      | Supports the drum, gearing and operating mechanism.   | No cracking, distortion, corrosion, loose covers or signs of movement.                        |
| <b>Handle / crank</b>          | Provides manual operating force.                      | Correct handle fitted, secure attachment, no bending, slipping or damaged grip.               |
| <b>Brake / locking device</b>  | Holds the load and prevents uncontrolled movement.    | Brake holds securely; locking device engages; no drift, creep or slipping.                    |
| <b>Drum</b>                    | Stores and winds rope, wire rope or belt.             | Correct winding, no crushing, bunching, damage, riding turns or poor fleet angle.             |
| <b>Rope / wire rope / belt</b> | Connects the winch to the load or moving component.   | Correct type and size; no broken wires, kinks, cuts, crushing, abrasion or corrosion.         |
| <b>Terminations</b>            | Secure the rope or wire rope at load and anchor ends. | Correct clips, thimbles, sockets, anchor points and manufacturer-approved termination method. |
| <b>Pulleys / guides</b>        | Change direction or guide the rope path.              | Free movement, correct alignment, no wear, groove damage or insecure fixings.                 |
| <b>Mounting / support</b>      | Transfers load into the building, structure or frame. | Suitable fixings, no movement, cracking, corrosion, loose bolts or unsupported load path.     |

## 5. Regulatory and good-practice context

The exact legal status depends on the installed use and local risk assessment. The following summary is intended to support classification in the portal and should be reviewed against the actual installation.

|                                 |                                                                                                                                                                                                                                           |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LOLER 1998</b>               | Where used for lifting at work, a manual winch is lifting equipment. Key issues include strength and stability, safe working load marking, planning of lifting operations, inspection and thorough examination by a competent person.     |
| <b>PUWER 1998</b>               | A manual winch is work equipment. It must be suitable for its intended use, maintained in a safe condition, inspected where deterioration could create risk, and used only by people with suitable information, instruction and training. |
| <b>BS EN 17206</b>              | Relevant where the winch forms part of entertainment machinery or a stage / production lifting system. It supports design, safety, control, verification, documentation and inspection principles.                                        |
| <b>ABTT Technical Standards</b> | Recognised theatre-industry good practice should be followed for installation, inspection, operation, maintenance, competence and record keeping in places of entertainment.                                                              |

### Records to retain

Retain manufacturer instructions, certificates, design/loading information, commissioning records, inspection reports, maintenance records, defect reports and details of any repair or modification.

## 6. Typical inspection and functional checks

| Check area            | What to check                                                                                                   | Result            | Notes / action |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|-------------------|----------------|
| <b>Identification</b> | Check the equipment identity, SWL / WLL, asset number and relevant markings are present and legible.            | Pass / Fail / N/A |                |
| <b>Frame / casing</b> | Inspect for cracks, distortion, corrosion, impact damage, loose covers, missing fasteners or signs of movement. | Pass / Fail / N/A |                |

| Check area                     | What to check                                                                                                                | Result            | Notes / action |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|
| <b>Mounting</b>                | Check wall, floor, frame or structural fixings for security, corrosion, damage or movement.                                  | Pass / Fail / N/A |                |
| <b>Handle / crank</b>          | Check handle condition, secure fitting, correct operation and absence of slipping or excessive play.                         | Pass / Fail / N/A |                |
| <b>Brake / lock</b>            | Check that the brake or locking device holds the load securely and releases only as intended.                                | Pass / Fail / N/A |                |
| <b>Drum</b>                    | Check rope winding, anchorage, fleet angle, drum damage, flange damage, bunching or crossed turns.                           | Pass / Fail / N/A |                |
| <b>Rope / wire rope / belt</b> | Inspect for broken wires, kinks, crushing, cuts, abrasion, corrosion, contamination or other deterioration.                  | Pass / Fail / N/A |                |
| <b>Terminations</b>            | Check sockets, clamps, thimbles, anchor points and dead-end arrangements for correct fitting and condition.                  | Pass / Fail / N/A |                |
| <b>Pulleys / guides</b>        | Check sheaves, diverters and guides for free rotation, wear, alignment and secure fixing.                                    | Pass / Fail / N/A |                |
| <b>Operation</b>               | Operate where safe to do so and check for smooth movement, abnormal noise, snatching, binding, creep or uncontrolled motion. | Pass / Fail / N/A |                |
| <b>Load path</b>               | Confirm the complete load path is understood and remains suitable for the current use.                                       | Pass / Fail / N/A |                |
| <b>Documentation</b>           | Confirm inspection, maintenance and defect records are available and up to date.                                             | Pass / Fail / N/A |                |

## 7. Remove from service if any of the following are found

- Unknown or illegible SWL / WLL where the rating cannot be verified.
- Brake, pawl, ratchet, locking device or hold function is unreliable.

- Damaged, kinked, crushed, badly corroded or incorrectly terminated rope / wire rope / belt.
- Loose, cracked, corroded or moving mountings or supporting structure.
- Cracked, distorted, badly worn or modified winch body, drum, gearing or handle.
- Abnormal operation such as slipping, drift, uncontrolled movement, grinding, jamming or sudden snatching.
- Evidence of overload, shock loading, unauthorised modification, poor repair or missing parts.
- Inspection status or competent person approval is unclear.

## 8. Maintenance and record keeping

- Follow the manufacturer maintenance instructions and any site-specific maintenance plan.
- Lubrication, adjustment, rope replacement and brake work should only be completed by competent persons.
- Do not replace rope, wire rope, belt, terminations, handles, brakes or fixings with non-equivalent parts unless approved.
- Record defects, repairs, adjustments, parts replacement, inspection findings and return-to-service approval.
- After significant repair, alteration, reinstallation or suspected overload, obtain competent assessment before continued use.

## 9. Portal document details

|                                 |                                                                  |
|---------------------------------|------------------------------------------------------------------|
| <b>Portal equipment item</b>    | Manual Winch                                                     |
| <b>Recommended PDF filename</b> | manual_winch_technical_datasheet.pdf                             |
| <b>Recommended GitHub path</b>  | documents/manual_winch_technical_datasheet.pdf                   |
| <b>Document status</b>          | Generic template for review and editing                          |
| <b>Prepared for</b>             | Stage Electrics - Mechanical Theatre Equipment Compliance Portal |

### Final review before use

Before this datasheet is published or relied upon, update it with the actual manufacturer, model, rating, installed configuration, inspection interval and any site-specific limitations.